

Vig Removal Changes the Answer

A pre-specified calibration test of 15,351 archived NBA moneylines

Jacob Burdier · The Scholars' Academy · 2026

15,351

games

5

vig rules

1

pre-set test



THE QUESTION

Do archived NBA moneylines misstate how often favorites win, by enough to be both statistically detectable and worth betting on?

Narrow on purpose. Not "are betting markets efficient."
One population, one threshold, one test, all fixed before I looked at a single outcome.

Everything below was frozen before any outcome was seen.

Population

NBA regular season, 2007-08 to 2019-20

Slice

favorites above .70 vig-free probability

Estimate

the calibration gap, in percentage points

WHY MOST TESTS HERE CANNOT ANSWER IT

A 74-game bucket decides nothing

11%

chance of catching a gap big enough to bet on

Papers here routinely report a bucket of favorites priced near .85 with about 74 games in it. Run the power calculation and that test finds a break-even-sized gap 11 percent of the time.

This is not a small problem at the edges. It is the reason the published record disagrees with itself.

74 games

11%

6,840 games

>99.9%

the gap you would need

2.99 points

Three claims that are easy to mix up

Calibration

Do stated probabilities match how often those outcomes happen?

What this paper tests

Profitability

Does a betting rule make money after paying the quoted price?

Tested second, directly

Efficiency

Do prices absorb information so well that no strategy beats them?

Not claimed at all

A market can be well calibrated on average and still hold predictable patterns in particular situations. It can also carry errors too small to turn into profit once the margin is paid.

What was fixed before I saw any outcome

Confirmatory · fixed in the plan

- The .70 threshold
- The Poisson-binomial primary test
- Proportional vig removal
- Five probability buckets, Holm corrected
- Flat-stake returns
- The smallest effect worth detecting

Exploratory · added at review

- Season-by-season gaps
- Cluster-robust variance
- Block bootstraps
- The other four vig-removal rules
- Equivalence testing

Checked before it was analyzed

15,351

games analyzed

30,978

prices cross-checked

0

mismatches found

13

season counts verified

The archive passed through two other hands before it reached me, so it was audited first. A separate team-level extraction in the same source repository was compared value by value against the analysis file. Season totals were checked against the real NBA schedule, including the 990-game lockout year and the 971 games played before the March 2020 shutdown.

A hand-drawn 260-game spot check against the live original archive also matched, but that is a sanity check. The automated full-file comparison is the real audit.

Every game gets its own probability

Bucket tests pretend every game in a range shares one probability. They do not. A -450 favorite and a -240 favorite are different bets.

So each game is its own coin with its own weight, and the number of favorite wins follows a Poisson-binomial distribution.

One coin per game, not one coin per bucket.

Expected	5,489.45
Observed	5,529
Gap	+0.58 pp
z / p	1.22 / .223

exact, not a normal approximation

PRIMARY RESULT

+0.58

percentage points

Across 6,840 games, favorites above .70 won 80.83% of the time against a stated 80.26%.

Holds under every variance estimator

Poisson-binomial $p = .223$

Classical $p = .218$

By season (13) $p = .134$

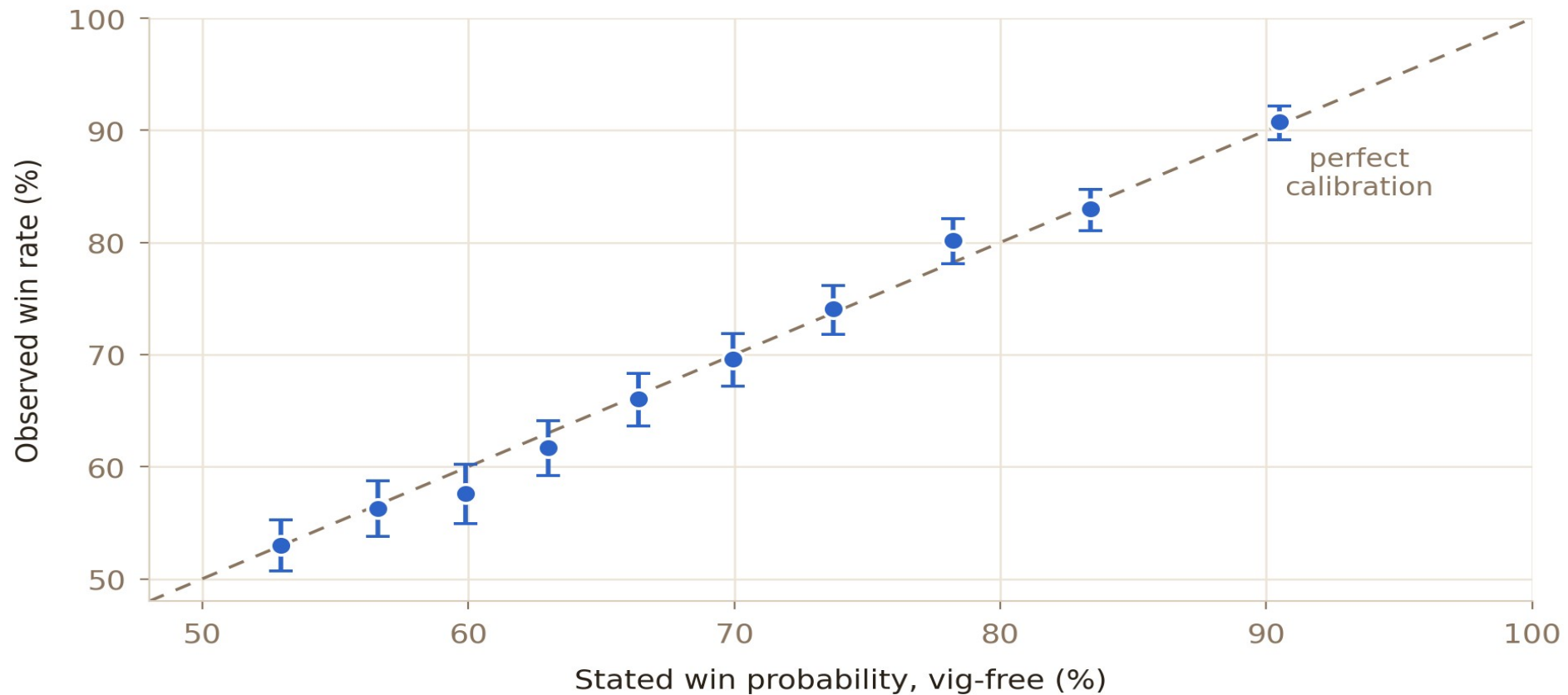
By favorite team (32) $p = .276$

By underdog team (33) $p = .246$

Two-way (13) $p = .224$

Point estimate is +0.58 in all six. Only the standard error moves.

Stated probability against what actually happened



Ten deciles of 15,351 games, 1,347 to 1,804 games each, with 95 percent Wilson intervals. The dashed line is perfect calibration.

"Not significant" is not the same as "small"

Failing to reject zero says the data are consistent with zero. It does not say the gap is too small to matter. Those are different claims, so the second one gets its own test.

Two one-sided tests

gap is below +2.99 $p = 4.7\text{e-}08$

gap is above -2.99 $p = 2.4\text{e-}13$

Both rejected. The gap is smaller than a bettor needs.

And the intervals agree on their own

Primary 95% upper bound **+1.51**

Team bootstrap upper bound **+1.57**

What a bettor needs **+2.99**

How big does a gap have to be to be worth money?

$$\text{expected return} = \text{true probability} / \text{raw price} - 1$$

Set that to zero and solve. The portfolio breaks even on a gap weighted by $1 / \text{raw probability}$, not on a plain average.

The paper originally quoted the plain average. The exact answer is two hundredths of a point away.

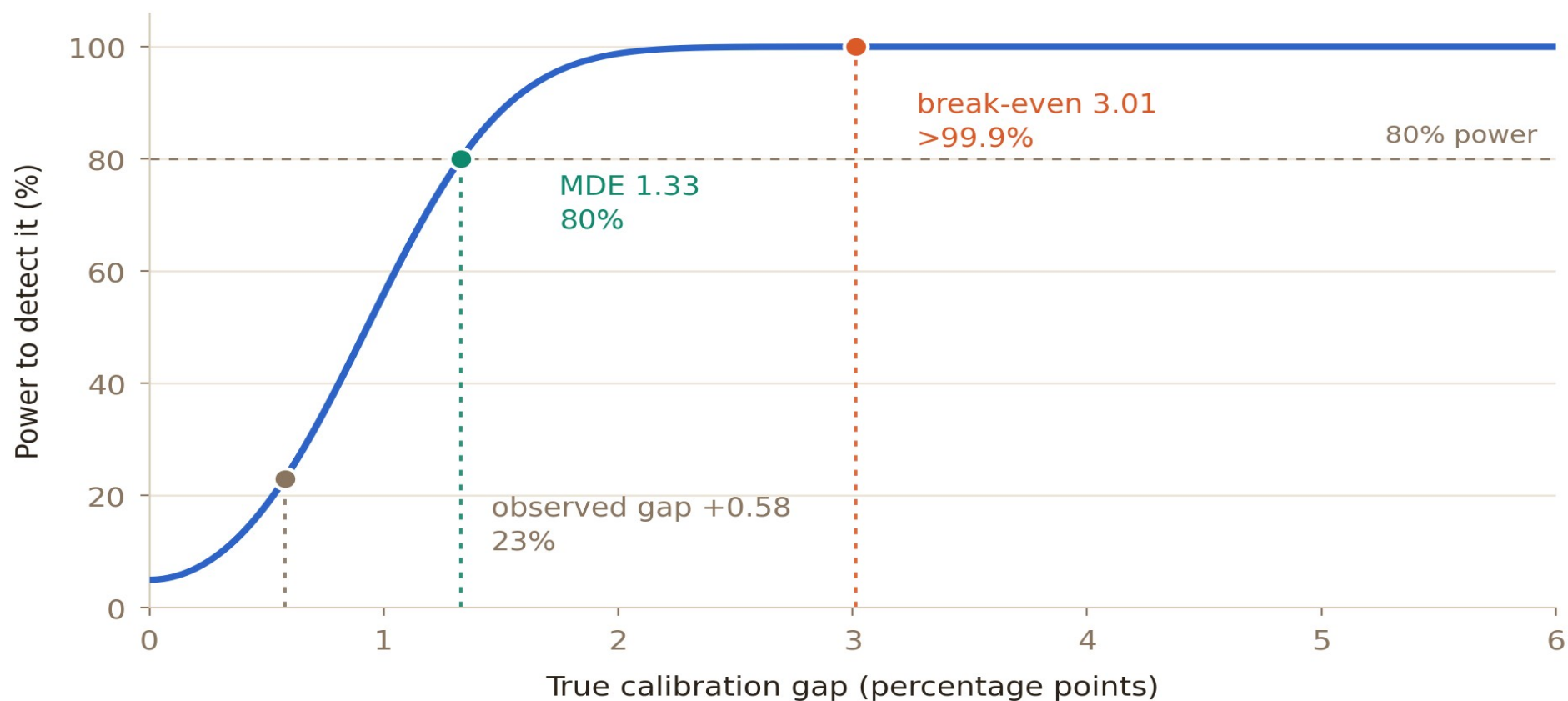
3.0126 pp

plain average, what the paper quoted

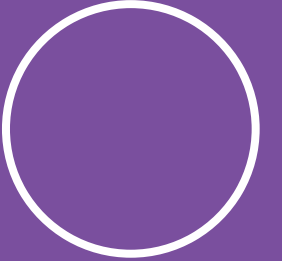
2.9941 pp

exact, what the tests now use

What size of gap could this design actually catch?



Above 99.9% power at the break-even margin, 80% at 1.33 points, about 11% for the 74-game bucket. No mark is taken from anyone else's paper.



So far, an unsurprising null.

Then I changed one line of code.

The bookmaker's margin has to come out of the price before you can test anything. There is no single correct way to take it out.

SAME 6,840 GAMES. ONLY THE RULE CHANGES.

The answer moves, and so does its sign

Vig-removal rule	Gap	p	Break-even	Verdict
None, raw prices	−2.43	<.001	—	margin artifact
Proportional (pre-set)	+0.58	.223	3.01	below the bar
Additive / Shin	−0.55	.231	1.88	below the bar
Constant odds ratio	−0.65	.157	1.78	below the bar
Power	−1.21	.008	1.22	below the bar

Each rule also picks its own games above .70, so an earlier version compared different rules on different samples. Holding the sample fixed is what this table does.

Three treatments of the same games

-2.43

raw prices

$p < .001$

+0.58

proportional

$p = .223$

-1.21

power

$p = .008$

Whether favorite bias appears at all, how big it is, and which way it points all turn on one modeling step.

Skip the margin and you measure the margin

-2.43

favorite bias, $p < .001$, testing against raw prices

Test observed win rates against raw implied probabilities and you get a large, highly significant favorite bias.

It is not better behavior. It is the bookmaker's cut, sitting in the numbers, being read as a result.

Several bucket-test designs still in use skip this step. The sign of the headline finding turns on it.

Shin's rule and the equal-margin rule are the same rule

Shin (1993) prices against a share of insider money. The equal-margin rule is a mechanical convention with no story behind it. For a two-outcome book they give identical answers, every time.

sum to one + difference held fixed → one possible pair

3.3e-16

difference preserved

2.8e-16

closed form

2.8e-16

Shin minus additive

2.2e-16

sum constraint

Largest error across 15,350 games: 3.3×10^{-16} . That is machine precision, not agreement to a few decimals.

Things this paper got wrong, on the record

A literature range with no source

An early draft cited favorite bias of 1.7 to 5.1 points. Neither number could be traced to any paper I cited. Both were removed rather than re-sourced.

A power claim doing the wrong job

The minimum detectable effect was being used as if it ruled small effects out. It does not. An equivalence test now carries that claim.

A table comparing different samples

Each vig rule selected its own games, so one comparison mixed the rule with sample composition. Fixing it retracted a claim about power normalization.

One command rebuilds every number

```
$ bash run_all.sh
```

1 verify the data

2 build the dataset

3 confirmatory tests

4 robustness

5 figures

6 vig rules and dependence

7 power curve

8 Shin proof

9 return simulation

10 equivalence and fixed sample

Two conclusions, one of them useful

For bettors

In these archived prices, over this period, under the rule fixed in advance, simply backing favorites or underdogs offered nothing worth having.

That is a narrow statement and it should stay narrow.

For researchers

The evidence for favorite bias is not robust to how the margin is removed. That dependence is invisible to a single-rule bucket test.

Reporting the sensitivity is not a caveat attached to the result. It is the result.

The title of this paper used to be a question about calibration. The data changed it into a question about method.

Limits, stated plainly

The freeze has no outside timestamp

The plan was fixed before I saw outcomes, but the record is in my own files. Pre-specified, not pre-registered.

One consensus line per game

No line shopping, no book identity, and the timing of each quote is unrecorded. Nothing here speaks to closing lines.

Thirteen clusters is not many

Season-level cluster-robust variance needs more groups than this to be trusted, so those rows are sensitivity checks.

The archive stops in March 2020

Everything predates most of the US legal expansion. No claim is made about what came after.

Questions? Holes in it?

The whole thing rebuilds from raw data with one command. I would rather it broke now than after someone cites it.

github.com/jacobburdier05/nba-moneyline-calibration

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